Module 3: The 70% Problem and Real Workflows

Structured workflows for partially correct output

Module 3: The 70% Problem and Real Workflows

Primary sources: Osmani, Chapter 3; Thomas & Hunt, Topic 12: Tracer Bullets.

Learning Outcomes

- Describe the 70 percent problem.
- Apply structured workflows to guide generation.
- Compare naive prompting with structured approaches.
- Implement a basic feature using an AI-assisted workflow.

What Is the 70 Percent Problem?

AI output often looks close enough to create confidence but not close enough to trust.

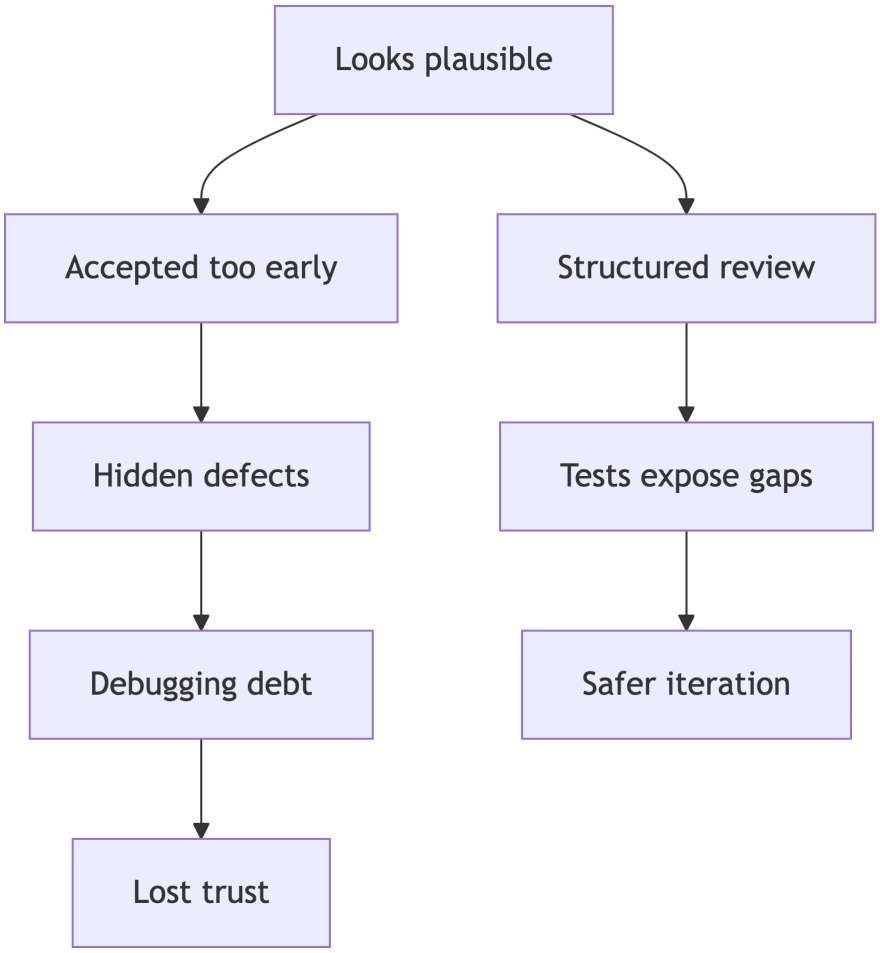
Osmani's core warning uses the "70 percent" idea as a heuristic: AI can often produce the patterned, visible part of a solution, while edge cases, architecture, maintainability, and production readiness remain human engineering work.

Common shape:

- Compiles, but misses edge cases.
- Solves the sample, but not the requirement.
- Uses plausible APIs incorrectly.

- Produces code that is hard to maintain.

The 70 Percent Trap



Why It Is Dangerous

Partially correct output can be worse than obviously wrong output.

- It saves time early.

- It shifts cost into debugging and review.
- It creates sunk-cost pressure.
- It can hide security or reliability defects.

Common failure modes include the "two steps back" loop and the demo-quality trap: an impressive happy path that collapses under real use.

Real Workflow Patterns

Osmani describes three useful AI-assisted workflow patterns.

- AI as first drafter: generate, then refine, refactor, and test.
- AI as pair programmer: work in tight feedback loops with frequent review.
- AI as validator: write code first, then use AI to critique, test, and improve it.

The pattern matters less than the discipline around it: review, test, commit small, and understand what you merge.

Workflow Over Heroics

Effective AI-assisted development uses a loop:

1. Define intent.
2. Ask for a small implementation.
3. Inspect the diff.
4. Run tests.
5. Refine prompt or code.
6. Document decisions.

Tracer Bullets

Thomas and Hunt recommend building thin, end-to-end slices that prove direction.

In AI-assisted SWE:

- Ask AI to help create the smallest complete path.
- Keep scope narrow enough to evaluate.
- Prefer inspectable progress over broad unfinished generation.
- Use the slice to get feedback under real project constraints.

Tracer code is not throwaway code. It is a lean but production-shaped skeleton that can grow as the system becomes clearer.

Golden Rules for the Remaining Work

Useful guardrails from Osmani Chapter 3:

- Be specific and clear about what you want.
- Validate AI output against your intent.
- Treat AI as a junior developer with supervision.
- Use AI to expand capability, not replace thinking.
- Isolate AI changes in version control.
- Do not merge code you do not understand.
- Document decisions, prompts, and useful patterns.

Bad AI Workflow

- Ask for a whole feature.
- Paste output without reading.
- Fix compile errors manually.
- Skip tests because it “seems right.”
- Document only the final result.

Better AI Workflow

- Ask for an implementation plan first.
- Generate one component at a time.
- Require tests before or alongside code.
- Compare output against criteria.
- Preserve rejected suggestions and rationale.

Lab 2 Development Connection

Eval-driven development starts this week.

Your lab goal is to make correctness measurable before relying on AI output.

Evaluation criteria should be:

- Specific.
- Runnable or inspectable.
- Connected to task intent.
- Useful for regression.

Practice Activity

Given a Java feature request, define:

- One happy-path test.
- Two edge-case tests.
- One invalid-input test.
- One maintainability criterion.

Then ask Copilot to implement only enough code to pass those checks.

Takeaways

- The last 30 percent is engineering work.
- Small slices expose uncertainty early.
- AI is most useful when surrounded by feedback loops.